# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Developed and maintained automated test suites for **Swift** applications on **iOS** and **macOS** platforms, ensuring comprehensive test coverage.
- Integrated **Large Language Models (LLMs)** into existing testing workflows to enhance automation capabilities and identify novel test cases.
- Designed and implemented test strategies utilizing the **XCTest framework**, focusing on functional, integration, and regression testing for core platform features.
- Collaborated with cross-functional teams to refine automation scripts and improve the efficiency of the release engineering pipeline.
- Participated in code reviews for automation code, providing constructive feedback to peers to maintain high standards of code quality.
- Analyzed and debugged complex issues within automated test runs, identifying root causes and implementing effective solutions.
- Containerized automation test environments using **Docker**, facilitating consistent execution across development and staging infrastructure.

## SKILLS

**Programming Languages** – Swift, Python, Java, SQL

**Testing Frameworks** – XCTest, JUnit

**AI/ML Technologies** – GenAI, LLMs, Machine Learning Model Training, Fine-tuning

**Platforms & Tools** – iOS, macOS, AWS, Docker, Jenkins, Git, Jira, Confluence

**Concepts** – Automated Testing, Release Engineering, Localization Testing, Internationalization Testing, Accessibility Testing, Agile Methodologies

## PROJECTS

### Swift Automation for macOS Application

- Developed a suite of automated tests for a macOS application using **Swift** and the **XCTest framework**, covering user interface interactions and core functionalities.
- Integrated **GenAI** capabilities to generate diverse test data scenarios, improving the detection of edge cases in application logic.
- Refactored existing test scripts to enhance execution speed by **18%** and improve maintainability.

### LLM-Powered Test Data Generation

- Built a Python-based service to interact with **Large Language Models (LLMs)** for generating realistic and contextually relevant test data for localization testing.
- Fine-tuned an LLM on sample localization data to improve the accuracy and relevance of generated strings, reducing manual data creation effort by **40%**.
- Designed an API endpoint using **FastAPI** to serve generated test data to downstream automation pipelines.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)